Coding Basics ⌨

Cadette Coding for Good Badge — Girls Build Bots

Presented by your robotics mentors

Code runs your day

Trace one ordinary hour:

What happens	The code behind it
Your alarm goes off	a **scheduled event**
Your playlist shuffles	a **randomizing algorithm**
The thermostat kicks on	a **conditional** — *if below 68, heat on*

Here’s the kicker

Someone wrote every one of those decisions.
Coders decide how technology behaves.

Today you step from *user* to *builder*.

The four tools 🧰

Tool	What it does	Physical demo
Event	reacts to a trigger	when the door opens → wave
Conditional	makes a decision	if I say “blue,” stand up
Variable	remembers a value	a running count on the board
Loop	repeats	repeat until told to stop

Pseudocode: idea → program 

Let's write one line for each, together.

```
on button pressed:
    if time < best_time:
        best_time = time
```

Pseudocode is the bridge. Plain language, program shape.

Build: Reaction-Time Tester

A real tool, not a toy — reaction time is what sports scientists measure.

1. **Event** — on button press, start the round
2. **Variable** — a timer counts up; a score stores your best
3. **Conditional** — pressed too early → "Too soon!"; new best → update
4. **Loop** — play rounds forever, tracking the record
5. **Stretch** — two-player mode with two variables

All four tools. One program. ✅

Every one of you leaves with something that actually works.

Code Review Swap 👥

Swap laptops. On a sticky note, answer:

1. **What does this program do?**
2. Point to a **conditional**. Point to a **variable**.
3. **One thing you'd improve.**

Then discuss — and make one improvement to your own.

This is the job

Every professional software team on Earth does exactly this.

Feedback given and taken seriously is the most professional twenty minutes of your day.

Robot moment 🤖

Where the four tools live in our robot code:

- **Events** → sensor triggers
- **Conditionals** → if-statements steering autonomous mode
- **Variables** → tracking game state
- **Loop** → the main control loop

Ask us about the time one line bricked our robot at competition.

Closing question

Which of the four tools would you teach a friend first — and how?

What you accomplished ✅

- Identified code making decisions in everyday technology
- Explained events, conditionals, variables, and loops with examples
- Built a working program using all four
- Read a peer’s code and gave real review feedback

Level up 🚀

- **arcade.makecode.com** — build the two-player stretch mode
- Write **pseudocode** for something you use daily, then build it
- Run a **code review** on a friend's project — the habit is the skill
- Ask us what programming a competition robot is actually like

📣 Share Your Impact

Troop leaders & robotics teams — if you used this curriculum, please:

1. **Credit us** — the curriculum comes from **Girls Build Bots**
2. **Tag us** — **@girlsbuildbots** on Instagram
3. **Report your results** — email **girlsbuildbots@gmail.com** with how many girls took part, how many badges were earned, and how it went

This is how the program grows — and how the next troop and team get better materials.